

Data Reduction

L 17 to L 20

3.4 Data reduction

Data reduction is crucial in machine learning because it

- ✓ Enhances efficiency,
 - ✓ performance, and
 - ✓ interpretability
- This will be mitigating the risks of **overfitting** and **computational complexity**, especially when dealing with **large datasets**. It **simplifies complex data**, making it **easier to analyze, visualize, and build models**.
 - Complex data analysis and mining on huge amounts of data can take a long time, making such analysis impractical or infeasible.
 - Data reduction techniques can be applied to obtain a **reduced representation** of the data set that is much smaller in volume yet closely maintains the integrity of the original data.
 - Mining on the reduced data set should be more efficient yet produce the same (or almost the same) analytical results.

3.4.1 Overview of Data Reduction Strategies

- Data reduction strategies include,
 - *dimensionality reduction*
 - *numerosity reduction*, and
 - *data compression*.

Dimensionality reduction :

- It is the process of reducing the number of random variables or attributes under consideration.
- Dimensionality reduction methods include **wavelet transforms** and **principal components analysis**, which transform or project the original data onto a smaller space.
- **Attribute subset selection** is a method of dimensionality reduction in which irrelevant, weakly relevant, or redundant attributes or dimensions are detected and removed.

3.4.1 Overview of Data Reduction Strategies

Numerosity Reduction:

- This techniques replace the original data volume by alternative, smaller forms of data representation.
- These techniques may be parametric or nonparametric.

Parametric methods: A model is used to estimate the data, so that only the data parameters need to be stored, instead of the actual data. (Outliers may also be stored.)

Example : Regression and log-linear models

Nonparametric methods :

- For storing reduced representations of the data include histograms, clustering , sampling , and data cube aggregation.

3.4.1 Overview of Data Reduction Strategies

Data compression ;

- Here transformations are applied so as to obtain a reduced or “compressed” representation of the original data.
- If the original data can be reconstructed from the compressed data without any information loss, the data reduction is called lossless.
- If, instead, we can reconstruct only an approximation of the original data, then the data reduction is called **lossy** **(some of the original data is discarded during the compression process to achieve a smaller file size)**
- **Dimensionality reduction** and **numerosity reduction techniques** can also be considered forms of data compression.
- The computational time spent on data reduction should not outweigh or “erase” the time saved by mining on a reduced data set size.

Principal Component Analysis: Real-Life Analogy

Shadow Projection Analogy

Think of PCA like analyzing shadows of 3D objects:

3D Object: Complex dataset with many dimensions

Light Source Direction: The perspective we choose to view the data

Shadow on Wall: Projection onto fewer dimensions (principal components)

Most Informative Shadow: The projection that preserves maximum variance



Just as we can learn about a 3D object from its shadow:

PCA finds the "best angle" to view high-dimensional data

The first principal component captures the most information

Each subsequent component captures the maximum remaining variance

We can often understand the data using just a few components

3.4.3 Principal Components Analysis

- It is a method of dimensionality reduction.
- Suppose the data to be reduced consist of tuples or data vectors described by **n** attributes or dimensions.
- Principal components analysis (PCA; also called the Karhunen-Loeve, or K-L, method) searches for **k n-dimensional orthogonal vectors** that can best be used to represent the data, where $k \leq n$.
- The original data are thus projected onto a much smaller space, resulting in dimensionality reduction.
- Unlike attribute subset selection, which reduces the attribute set size by retaining a subset of the initial set of attributes, PCA “combines” the essence of attributes by creating an alternative, smaller set of variables.
- The initial data can then be projected onto this smaller set. PCA often reveals relationships that were not previously suspected and thereby allows interpretations that would not ordinarily result.

3.4.3 Principal Components Analysis

The **basic procedure** is as follows:

1. The **input data are normalized**, so that each attribute **falls within the same range**. This step helps ensure that attributes with large domains will not dominate attributes with smaller domains.
2. PCA computes **k orthonormal vectors** that provide a basis for the normalized input data. These are unit vectors that each point in a direction perpendicular to the others.
 - These vectors are referred to as the principal components. The input data are a linear combination of the principal components.
3. The principal components are sorted in order of decreasing “significance” or strength. The principal components essentially serve as a new set of axes for the data, providing important information about variance.
 - That is, the sorted axes are such that the first axis shows the most variance among the data, the second axis shows the next highest variance, and so on.

3.4.3 Principal Components Analysis

- For example, Figure 3.5 shows the first two principal components, Y_1 and Y_2 , for the given set of data originally mapped to the axes X_1 and X_2 . This information helps identify groups or patterns within the data.

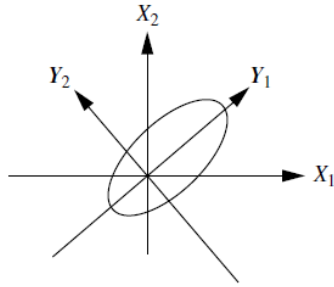


Figure 3.5 Principal components analysis. Y_1 and Y_2 are the first two principal components for the given data.

- Because the components are sorted in decreasing order of “significance,” the data size can be reduced by eliminating the weaker components, that is, those with low variance.
- Using the strongest principal components, it should be possible to reconstruct a good approximation of the original data.
- PCA can be applied to ordered and unordered attributes, and can handle sparse data and skewed data.
- Multidimensional data of more than two dimensions can be handled by reducing the problem to two dimensions.
- Principal components may be used as inputs to multiple regression and cluster analysis. In comparison with wavelet transforms, PCA tends to be better at handling sparse data, whereas wavelet transforms are more suitable for data of high dimensionality.

Principal Component Analysis: Step-by-Step Process

PCA Implementation Process

1 Standardize the Data

Scale features to have mean=0 and variance=1 to ensure all features contribute equally.

2 Compute Covariance Matrix

Calculate how each variable correlates with every other variable.

3 Calculate Eigenvalues & Eigenvectors

Find the directions (eigenvectors) and magnitudes (eigenvalues) of maximum variance.

4 Sort & Select Components

Arrange eigenvalues in descending order and select top k eigenvectors.

5 Project Data

Transform original data onto the new subspace defined by principal components.

3.4.4 Attribute Subset Selection

- Data sets for analysis may contain hundreds of attributes, many of which may be irrelevant to the mining task or redundant.
- For example, if the task is to **classify customers** based on whether or not they are likely to purchase a popular new CD at AllElectronics when notified of a sale, attributes such as the customer's telephone number are likely to be irrelevant, unlike attributes such as age or music taste.
- Although it may be possible for a domain expert to pick out some of the useful attributes, this can be a difficult and time consuming task, especially when the data's behavior is not well known.
- Leaving out relevant attributes or keeping irrelevant attributes may be detrimental, causing confusion for the mining algorithm employed.
- This can result in discovered patterns of poor quality. In addition, the added volume of irrelevant or redundant attributes can slow down the mining process.

3.4.4 Attribute Subset Selection - In machine learning, attribute subset selection is known as *feature subset selection*.

- Attribute subset selection reduces the data set size by removing irrelevant or redundant attributes (or dimensions). The goal of attribute subset selection is to find a minimum set of attributes such that the resulting probability distribution of the data classes is as close as possible to the original distribution obtained using all attributes.
- Mining on a reduced set of attributes has an additional benefit: It reduces the number of attributes appearing in the discovered patterns, helping to make the patterns easier to understand.

How can we find a ‘good’ subset of the original attributes?

- ”For n attributes, there are 2^n possible subsets. An exhaustive search for the optimal subset of attributes can be prohibitively expensive, especially as n and the number of data classes increase.
- Therefore, heuristic methods that explore a reduced search space are commonly used for attribute subset selection.
- These methods are typically **greedy** in that, while searching through attribute space, they always make what looks to be the best choice at the time.
- Their strategy is to make a locally optimal choice in the hope that this will lead to a globally optimal solution. Such greedy methods are effective in practice and may come close to estimating an optimal solution.

3.4.4 Attribute Subset Selection

- The “best” (and “worst”) attributes are typically determined using tests of statistical significance, which assume that the attributes are independent of one another.
- Many other attribute evaluation measures can be used such as the information gain measure used in building decision trees for classification.
- Basic heuristic methods of attribute subset selection include the techniques that follow, some of which are illustrated in Figure 3.6.

Forward selection	Backward elimination	Decision tree induction
Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ Initial reduced set: $\{\}$ $\Rightarrow \{A_1\}$ $\Rightarrow \{A_1, A_4\}$ $\Rightarrow$ Reduced attribute set: $\{A_1, A_4, A_6\}$	Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ $\Rightarrow \{A_1, A_3, A_4, A_5, A_6\}$ $\Rightarrow \{A_1, A_4, A_5, A_6\}$ $\Rightarrow$ Reduced attribute set: $\{A_1, A_4, A_6\}$	Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ <pre>graph TD; A4["A4?"] -- Y --> A1["A1?"]; A4 -- N --> A6["A6?"]; A1 -- Y --> C1_1((Class 1)); A1 -- N --> C2_1((Class 2)); A6 -- Y --> C1_2((Class 1)); A6 -- N --> C2_2((Class 2));</pre> $\Rightarrow$ Reduced attribute set: $\{A_1, A_4, A_6\}$

Figure 3.6 Greedy (heuristic) methods for attribute subset selection.

3.4.4 Attribute Subset Selection

- 1. Stepwise forward selection:** The procedure starts with an empty set of attributes as the reduced set. The best of the original attributes is determined and added to the reduced set. At each subsequent iteration or step, the best of the remaining original attributes is added to the set.
- 2. Stepwise backward elimination:** The procedure starts with the full set of attributes. At each step, it removes the worst attribute remaining in the set.
- 3. Combination of forward selection and backward elimination:** The stepwise forward selection and backward elimination methods can be combined so that, at each step, the procedure selects the best attribute and removes the worst from among the remaining attributes.
- 4. Decision tree induction:** Decision tree algorithms (e.g., ID3, C4.5, and CART) were originally intended for classification. Decision tree induction constructs a flowchart like structure where each internal (nonleaf) node denotes a test on an attribute, each branch corresponds to an outcome of the test, and each external (leaf) node denotes a class prediction. At each node, the algorithm chooses the “best” attribute to partition the data into individual classes.

3.4.4 Attribute Subset Selection

- When decision tree induction is used for attribute subset selection, a tree is constructed from the given data. All attributes that do not appear in the tree are assumed to be irrelevant.
- The set of attributes appearing in the tree form the reduced subset of attributes.
- The stopping criteria for the methods may vary. The procedure may employ a threshold on the measure used to determine when to stop the attribute selection process.
- In some cases, we may want to create new attributes based on others. Such **attribute construction** can help improve accuracy and understanding of structure in high dimensional data.
- For example, we may wish to add the attribute *area* based on the attributes *height* and *width*. By combining attributes, attribute construction can discover missing information about the relationships between data attributes that can be useful for knowledge discovery.

3.4.5 Regression and Log-Linear Models: Parametric Data Reduction

- Regression and log-linear models can be used to approximate the given data. In (simple) linear regression, the data are modeled to fit a straight line.
- For example, a random variable, y (called a response variable), can be modeled as a linear function of another random variable, x (called a predictor variable), with the equation

$$y = wx + b,$$

where the variance of y is assumed to be constant. In the context of data mining, x and y are numeric database attributes.

- The coefficients, w and b (called *regression coefficients*), specify the slope of the line and the y -intercept, respectively.
- These coefficients can be solved for by the method of least squares, which minimizes the error between the actual line separating the data and the estimate of the line.
- Multiple linear regression is an extension of (simple) linear regression, which allows a response variable, y , to be modeled as a linear function of two or more predictor variables.

3.4.5 Regression and Log-Linear Models: Parametric Data Reduction

- **Log-linear models** approximate discrete multidimensional probability distributions.
- Given a set of tuples in n dimensions (e.g., described by n attributes), we can consider each tuple as a point in an n -dimensional space.
- Log-linear models can be used to estimate the probability of each point in a multidimensional space for a set of discretized attributes, based on a smaller subset of dimensional combinations.
- This allows a higher-dimensional data space to be constructed from lower-dimensional spaces.
- Log-linear models are therefore also useful for dimensionality reduction (since the lower-dimensional points together typically occupy less space than the original data points) and data smoothing (since aggregate estimates in the lower-dimensional space are less subject to sampling variations than the estimates in the higher-dimensional space).

3.4.5 Regression and Log-Linear Models: Parametric Data Reduction

- Regression and log-linear models can both be used on sparse data, although their application may be limited.
- While both methods can handle skewed data, regression does exceptionally well.
- Regression can be computationally intensive when applied to high-dimensional data, whereas log-linear models show good scalability for up to 10 or so dimensions.